Machine Learning 101 for Financial Trading

FSE-DAT-3019-1.1 | Data Analytics and Computational Modelling

Sector: Financial Services | Proficiency Level 3

LESSON PLAN

Tertiary Infotech Academy Pte Ltd

UEN 201200696W

Version 2.7 | 23 October 2026

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Version History

Version No	Date Effected	Change Update	Approved by
2.7	23 October 2026	Assessment papers rebuilt in the WSQ format for digital marking: cover page; page 2 with the trainee's name, NRIC, date and signature, the Instructions to Candidate and the Grading block with the assessor's name, NRIC, date and signature; questions from page 3, each with an answer box. Practical Performance redesigned around the labs, all in the Trading Simulator and TradingView: Task 1 hypothesis testing (Lab 4), Task 2 technical indicators and backtesting (Labs 5-7, 9), Task 3 machine-learning model and A/B testing (Labs 11, 13) and Task 4 the multi-agent desk with human approval, forward test, diagnosis and report (Labs 16, 18-20); marking points and the Assessor Key rewritten. Written Assessment Q6 (pairs trading) and Q9 (forms of data and turning headlines into data) now follow the slides.	Dr Alfred Ang
2.6	23 October 2026	Lab 17 (TradingView webhook) explained for learners: a new 'What This Lab Is For' section (what a webhook is, that the simulator supplies the URL, and the checks every alert passes before an order); Step 1 explains each part of the webhook card and alert message, Step 2 explains each of the seven self-test alerts and its HTTP code, and Step 3 gives the full real-alert walk-through (ngrok tunnel, webhook URL, chart, Pine strategy, alert dialog Condition / Message / Notifications, a quick Price-alert demo and where to watch the alert arrive). The Lab 17 screenshot now shows the TradingView webhook card with the 7/7 self-test result. Every lab in the deck is more visual: a 'Step by Step' slide (where to click, what to do, what you should see) and a 'How to Read the Screen' slide with numbered, colour-coded callouts on the simulator screenshot and a one-line takeaway; Lab 13 also gets a 'How to Reach the A/B Decision' slide (paired t-test, bootstrap Sharpe CI, Bonferroni alpha).	Dr Alfred Ang
2.5	23 October 2026	Topic 3 (K2): four slides added after 'Families of Learning Algorithms' explaining supervised, unsupervised and reinforcement learning - one slide each with a concept chart (SPY data for supervised and unsupervised, an agent-environment loop diagram for reinforcement), then a side-by-side comparison. Learner Guide section 3.1 and the Facilitator Guide Topic 3 teaching points cover the same content. Trading Simulator screenshots updated to the current app, whose Research tabs now follow the lab order: Strategy search and Walk-forward (Lab 10), ML models (Lab 11), Regimes (Lab 12), A/B test (Lab 13), Pairs screen (Lab 14), the multi-agent desk (Lab 20) and the 14-stage Pipeline Map (Getting Started); Lab 8 and Lab 9 screenshot captions aligned with the lab steps.	Dr Alfred Ang
2.4	22 October 2026	Practical Performance question paper simplified: one answer box per task, in which the candidate answers every part of the task and pastes its screenshot - the hypothesis test result in Task 1 and a TradingView Strategy Tester backtest of the Bollinger Bands strategy on XLF in Task 2.	Dr Alfred Ang

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2.3	22 October 2026	Assessment papers: every Practical Performance sub-question now has its own answer box, with screenshot boxes naming the lab where the step was practised, and the Written Assessment's ruled lines are replaced by answer boxes, so candidates type or paste their answers into the paper and upload it to the LMS. Assessment Plan Annex A wording aligned with the Written Assessment paper.	Dr Alfred Ang
2.2	22 October 2026	Screenshots added to Labs 7, 8 and 9: a Bollinger Band mean-reversion strategy in TradingView with the Paper Trading order panel (Lab 7), the simulator's paper order ticket and its passed pre-trade risk checks and fill (Lab 8), and Performance Analytics comparing a strategy with SPY buy-and-hold (Lab 9). Slide references in the Lesson Plan, Learner Guide and Facilitator Guide updated to the new slide numbers.	Dr Alfred Ang
2.1	21 October 2026	Low-code delivery. Every lab now runs in the Trading Simulator without coding, and the Practical Performance is done by configuring and running a provided trading program, with the same scenario, tasks, ability coverage and marking. A Prompt Library provides one AI prompt per lab, with one-click links, for learners who want code generated for them. Getting Started screenshots added (simulator Pipeline Map and Integrations, Alpaca paper dashboard, TradingView chart). Version History added to every document. Topic 2 slides added on mean reversion (the simulator's rules, mean reversion versus trend following, the main risk and safeguards), on the RSI look-back period (RSI(2) versus RSI(14)) and on long and short positions.	Dr Alfred Ang
2.0	18 October 2026	Major revision, focused entirely on quant and algorithmic stock trading. Courseware rebuilt and re-platformed to Alpaca (API-first broker, paper trading, 24/5 US stock trading) and TradingView (charting, Pine Script v6, Strategy Tester, Paper Trading) with Python. Four topics: quant success stories and the evolution of quant from Thorp and Renaissance through statistical arbitrage to high-frequency trading, the paradigm shift from machine learning to LLMs and its impact on quant, and AI agents versus agentic AI in trading; the classification of trading algorithms, indicators, strategies, a TradingView versus Alpaca comparison, order placement and metric definitions; machine learning models, statistical modelling, A/B testing and pairs trading as a hedging algorithm; and an end-to-end, automated algorithmic trading pipeline. 'How the algorithm works' slides explain each algorithm on real market data. Twenty hands-on labs build one pipeline stage by stage, each with an instruction PDF containing step-by-step AI prompts, and include live paper trading of a strategy. Assessment: Written Assessment (short-answer questions, K1-K9) and Practical Performance (four tasks, A1-A10) with a standalone SMART marking rubric. Facilitator Guide added. Fee restated in SGD exclusive of GST, optimum class size stated, provider name updated to Tertiary Infotech Academy Pte Ltd.	Dr Alfred Ang
1.1	22 December 2022	Multiple choice questions removed; practical test retained as the sole assessment instrument.	Dr Alfred Ang
1.0	10 June 2022	Initial release for pilot launch, aligned to Data Analytics and Computational Modelling (Proficiency Level 3).	Dr Alfred Ang

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1. Course Information

Item	Details
Course Title	Machine Learning 101 for Financial Trading
TSC Code and Title	FSE-DAT-3019-1.1 Data Analytics and Computational Modelling
Sector	Financial Services
TSC Category	Data Analytics and Information Technology Management
Proficiency Level	3
Course Duration	16 hours over 2 training days (14 hours classroom training + 2 hours assessment)
Daily Schedule	0900 - 1830 (8 instructional hours per day, excluding two 15-minute tea breaks and a 1-hour lunch break)
Mode of Delivery	Instructor-led classroom training with hands-on practical laboratory exercises on the Alpaca paper-trading platform, TradingView and the low-code Trading Simulator
Software Platforms	Alpaca paper-trading account (Trading API and Market Data API, alpaca-py SDK); TradingView (charts, Pine Script v6, Strategy Tester, alerts); the course's low-code Trading Simulator (https://github.com/alfredang/algorithmic-trader-bot)
Assessment Methods	Written Assessment - Short Answer Questions (45 minutes) | Practical Performance (75 minutes)
Training Venue	12 Woodlands Square, #07-85/86/87, Woods Square Tower 1, Singapore 737715
Class Size	Optimum class size of 20 participants (maximum 20)
Trainer to Learner Ratio	1 : 20
Course Fee	SGD 800 per participant (excluding GST)

2. Target Audience and Entry Requirements

Target Audience

- Traders, dealers, investment and research analysts in banks, brokerages and asset managers who want to test and automate trading ideas with data and machine learning.
- Quantitative, risk and data analysts in financial institutions who support trading, treasury or investment functions.
- Finance executives and technology professionals moving into algorithmic trading, trading technology or AI roles in financial services.
- Individual investors working in financial services who want a disciplined, evidence-based approach to systematic trading.

Entry Requirements

Knowledge and Skills

- Able to read, write, speak and listen in English at a proficiency level equivalent to the Employability Skills Workplace Literacy Level 5.
- Able to manipulate numbers at a proficiency level equivalent to the Employability Skills Workplace Numeracy Level 5.

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- Basic computer literacy; no programming is required - every lab runs in a low-code trading simulator.
- Basic understanding of financial markets (stocks, ETFs, buying and selling).

Experience

- Preferably 1 year of working experience in financial services or a related role involving data, investment or trading.

Equipment

- A laptop (macOS, Windows or Linux) with a modern web browser and internet access, able to run the free course Trading Simulator (installation is guided in Lab 1).
- A free Alpaca paper-trading account and a free TradingView account, created in class; no real money is used at any point.

3. Learning Outcomes

On completion of this course, participants will be able to:

LO	Learning Outcome	TSC Reference
LO1	Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services.	A1, A6, K1, K9
LO2	Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements.	A2, A3, K3, K5, K7
LO3	Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses.	A4, A5, K2, K4, K6
LO4	Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly.	A7, A8, A9, A10, K7, K8

4. TSC Alignment

This course is fully aligned to FSE-DAT-3019-1.1 Data Analytics and Computational Modelling at Proficiency Level 3: "Identify and utilise appropriate statistical algorithms and data models to perform data analyses, test hypotheses and derive inferences, patterns or solutions from data". All 10 ability statements and all 9 knowledge statements of the TSC are delivered and assessed.

Abilities

Ref	Ability Statement	Delivered In	Assessed By
A1	Develop hypotheses for implementing data analytics and computational modelling activities	Topic 1	PP Task 1
A2	Identify appropriate statistical algorithms and data models to test hypotheses and/or theories	Topics 2, 3	PP Task 2

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Ref	Ability Statement	Delivered In	Assessed By
A3	Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements	Topics 1, 2	PP Task 2
A4	Conduct statistical modelling of data to derive patterns and solutions	Topic 3	PP Task 3
A5	Perform A/B testing based on different hypotheses	Topic 3	PP Task 3
A6	Identify possible big data applications	Topic 1	PP Task 1
A7	Perform coding and configuration of software agents or programs based on selected models or algorithms	Topic 4	PP Task 4
A8	Conduct tests on the actions taken and outcomes to assess effectiveness of the models	Topic 4	PP Task 4
A9	Diagnose unintended outcomes produced by statistical models	Topic 4	PP Task 4
A10	Draw relevant trends and insights from data analysis to support decisions and report accordingly	Topic 4	PP Task 4

Knowledge

Ref	Knowledge Statement	Delivered In	Assessed By
K1	Data analytics and modelling business use cases	Topics 1, 4	WA Q1
K2	Types of statistical analyses, data models, algorithms and advanced computational methods	Topic 3	WA Q2
K3	Range and application of various statistical methods and algorithms	Topic 2	WA Q3
K4	Range and application of various types of data models	Topic 3	WA Q4
K5	Usage of analytics platforms and tools	Topics 1, 2, 4	WA Q5
K6	Statistical modelling techniques	Topic 3	WA Q6
K7	Coding languages for programming of algorithms and signals	Topics 2, 4	WA Q7
K8	Potential reasons for unintended outcomes	Topic 4	WA Q8
K9	Forms of data within financial services	Topic 1	WA Q9

5. Course Structure and Duration by Topic

Topic	Title	Day	Labs	TSC Ref	Duration	Slides
Topic 1	Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents	Day 1	Labs 1-4	A1, A6, K1, K9	3 hours	35 - 80
Topic 2	Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca	Day 1	Labs 5-9	A2, A3, K3, K5, K7	4.25 hours	81 - 151
Topic 3	Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals	Day 2	Labs 10-14	A4, A5, K2, K4, K6	2.5 hours	152 - 191
Topic 4	Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis	Day 2	Labs 15-20	A7, A8, A9, A10, K7, K8	3.25 hours	192 - 222
Admin	Course overview, ground rules, daily recaps	1, 2	-	-	1 hour	1 - 34
Assessment	Written Assessment and Practical Performance	2	-	K1-K9, A1-A10	2 hours	-
TOTAL					16 hours	

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Topic hours include the trainer-led content and the hands-on labs for that topic. Training 14 hours + assessment 2 hours = 16 hours, excluding tea and lunch breaks.

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Hands-on labs

Lab	Title	Topic	TSC Ref	Duration	Slide
1	Pipeline Skeleton, Configuration and the Alpaca Paper Account	1	K5	20 min	67
2	Data-Validation Stage: Cross-Check Pipeline Bars against TradingView	1	A3, K5, K9	20 min	70
3	Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality	1	A6, K9	20 min	73
4	Hypothesis Stage: from a Written Card to an Automated Statistical Test	1	A1, A6, K1	15 min	77
5	Indicator Stage: Pine Script v6 Indicators and their Python Twins	2	A3, K3, K7	25 min	128
6	Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation	2	A2, A3, K3, K7	30 min	132
7	Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals	2	A2, K3, K7	25 min	137
8	Execution Stage: an Order Manager that Trades the Paper Account in Any Session	2	A3, K5	30 min	141
9	Evaluation Stage: Performance and Risk Metrics with a Benchmark Table	2	A3, K3	25 min	146
10	Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay	3	A2, A4, K2	20 min	174
11	ML-Signal Stage: a Walk-Forward Classifier from Probability to Position	3	A4, K4, K6	20 min	178
12	Regime-Filter Stage: PCA and K-Means as a Gate on the Signals	3	A4, K4, K6	15 min	181
13	A/B-Test Stage: a Statistical Decision before Capital Moves	3	A5, K2	15 min	184
14	Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve	3	A4, K6	20 min	188
15	Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even	4	A7, A8, K7	25 min	204
16	Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading	4	A7, A8, K7	25 min	207
17	Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution	4	A7, K5, K7	20 min	210
18	Diagnostics Stage: Automated Bias Checks before Go-Live	4	A9, K8	20 min	213
19	Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report	4	A8, A10	25 min	216
20	Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval	4	A7, A10, K1	20 min	219

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6. Daily Training Schedule

Each training day comprises 8 instructional hours from 0900 to 1830, excluding two 15-minute tea breaks and a 1-hour lunch break. The 2-hour final assessment (45-minute Written Assessment and 75-minute Practical Performance) is conducted at the end of Day 2.

Day 1

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
09:00 - 09:30	Course Overview, Trainer Introduction and Ground Rules Course objectives, TSC alignment, assessment overview, ground rules and an icebreaker on how participants use data in trading or investment today.	-	Trainer presentation; Slides; Learner Guide	-
09:30 - 10:45	T1.1-1.4 Quant Success Stories, the Evolution of Quant, the ML-to-LLM Shift, and Agentic AI versus AI Agents Quant success stories (Thorp, Renaissance); the evolution of quant to high-frequency trading and the six paradigm shifts; machine learning to LLMs and the impact on quant; agentic AI versus AI agents and multi-agent trading systems.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide	K1, K9
10:45 - 11:00	Tea Break			
11:00 - 11:30	T1.5-1.6 Forms of Financial Data, Big Data Applications and Trading Hypotheses Market, reference, fundamental and alternative data; the four Vs; big data applications; writing a testable hypothesis.	LO1	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A1, A6, K1, K9
11:30 - 11:50	Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab01_Instructions.pdf	K5
11:50 - 12:10	Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView Prove the app's prices are right: compare its bars with TradingView's Data Window and explain adjusted vs unadjusted prices.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab02_Instructions.pdf	A3, K5, K9
12:10 - 12:30	Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality Ingest bars from Alpaca / Twelve Data / yfinance (or the demo files), run the data-quality checks and score news sentiment.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab03_Instructions.pdf	A6, K9
12:30 - 13:30	Lunch Break			
13:30 - 13:45	Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test Write H0 and H1 before looking at results and test them with a Welch t-test and a bootstrap CI.	LO1	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab04_Instructions.pdf	A1, A6, K1
13:45 - 14:30	T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies SMA, EMA, RSI, MACD, Bollinger Bands, ATR, VWAP, OBV; trend following, mean reversion, momentum, breakout, pairs trading and execution algorithms.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A2, K3

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
14:30 - 14:55	Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins Plot SMA/EMA, Bollinger Bands, RSI and MACD and export a Pine Script v6 strategy for TradingView.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab05_Instructions.pdf	A3, K3, K7
14:55 - 15:25	Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation Run the same MA-crossover in TradingView's Strategy Tester and in the app, reconcile the results, and try TradingView Paper Trading.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab06_Instructions.pdf	A2, A3, K3, K7
15:25 - 15:40	Tea Break			
15:40 - 16:05	Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals Pick and customise one strategy from each family and compare them after costs.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab07_Instructions.pdf	A2, K3, K7
16:05 - 16:50	T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement TradingView, Alpaca Trading and Market Data APIs, Python libraries; Pine Script and Python; order types, time in force, bracket orders and the order lifecycle.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A3, K5, K7
16:50 - 17:20	Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session Place market, limit, stop and bracket paper orders, including an extended-hours limit order, and follow each order's lifecycle.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab08_Instructions.pdf	A3, K5
17:20 - 17:50	T2.5 Performance and Risk Metric Definitions CAGR, volatility, Sharpe, Sortino, Calmar, maximum drawdown, win rate, profit factor, expectancy, exposure, turnover, alpha and beta.	LO2	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A3, K3
17:50 - 18:15	Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table Read the full metrics table and compare a strategy with buy-and-hold.	LO2	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab09_Instructions.pdf	A3, K3
18:15 - 18:30	Day 1 Recap and Q&A Consolidation of Topics 1 and 2; preview of Day 2.	-	Trainer presentation; Slides; Learner Guide	-

Total instructional time, Day 1: 8 hours (480 minutes), excluding the tea breaks and the 1-hour lunch break.

Day 2

Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
09:00 - 09:15	Day 1 Recap and Day 2 Overview Review of Topics 1 and 2; orientation to Topics 3 and 4 and to the assessment.	-	Trainer presentation; Slides; Learner Guide	-
09:15 - 09:40	T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models Descriptive to prescriptive analysis; supervised, unsupervised and reinforcement learning; strategy search; regression, trees, random forest, XGBoost, neural networks.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, K2, K4
09:40 - 10:00	Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay Grid-search SMA crossover parameters in-sample and measure how the ranking decays out of sample.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab10_Instructions.pdf	A2, A4, K2

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
10:00 - 10:20	Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position Train logistic regression, random forest and XGBoost walk-forward and trade P(up) > threshold.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab11_Instructions.pdf	A4, K4, K6
10:20 - 10:35	Tea Break			
10:35 - 10:55	T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques K-means, hierarchical clustering and PCA for regimes; returns and stationarity; time-ordered validation; feature engineering; regularisation.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, K4, K6
10:55 - 11:10	Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab12_Instructions.pdf	A4, K4, K6
11:10 - 11:25	T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm Hypotheses, Welch's t-test, bootstrap Sharpe difference, multiple testing; cointegration, hedge ratio, spread z-score and market neutrality.	LO3	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A4, A5, K2, K6
11:25 - 11:40	Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves Decide statistically between two variants before moving capital.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab13_Instructions.pdf	A5, K2
11:40 - 12:00	Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve Screen pairs for cointegration and backtest a market-neutral KO/PEP spread.	LO3	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab14_Instructions.pdf	A4, K6
12:00 - 12:30	T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs The eight-stage pipeline; configuration and pre-trade risk controls; next-bar execution; commission, spread, slippage and regulatory fees.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A7, A8, K7
12:30 - 13:30	Lunch Break			
13:30 - 13:55	Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even Run the same strategy under four cost models and find its break-even cost.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab15_Instructions.pdf	A7, A8, K7
13:55 - 14:05	T4.3 Forward Testing with Automated Paper Trading The bot loop, market clock, risk checks, order status and logging.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A7, A8
14:05 - 14:30	Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading Create a bot, run it on a schedule, watch orders, positions and P&L live, and halt it with the kill switch.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab16_Instructions.pdf	A7, A8, K7
14:30 - 14:50	Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution Send TradingView alerts to the app's webhook and see them pass validation, signal confirmation and risk into paper orders.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab17_Instructions.pdf	A7, K5, K7
14:50 - 15:00	T4.4 Potential Reasons for Unintended Outcomes Overfitting, look-ahead and survivorship bias, cost drag, regime change, data and software errors; the diagnosis method.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A9, K8

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Time	Topic / Activity	Learning Outcome	Delivery Method & Resources	TSC Ref
15:00 - 15:20	Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab18_Instructions.pdf	A9, K8
15:20 - 15:35	Tea Break			
15:35 - 15:45	T4.5 Performance Analysis, Insights and Reporting Tearsheets, backtest versus forward test, insights and the recommendation.	LO4	Trainer presentation, discussion and worked examples; Slides; Learner Guide	A10
15:45 - 16:10	Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report Produce a one-page tearsheet and explain the gap between backtest and forward test.	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab19_Instructions.pdf	A8, A10
16:10 - 16:30	Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval Run the whole pipeline end to end, get a human-approved proposal from the multi-agent desk, switch on the always-on agent, and create, backtest and forward-test your own strategy (from a template or written for you by AI).	LO4	Hands-on lab in the Trading Simulator; Learner Guide; lab manual Lab20_Instructions.pdf	A7, A10, K1
16:30 - 17:15	Written Assessment (Short Answer Questions) 9 open-ended short-answer questions covering K1 to K9.	All LOs	Individual assessment; Assessment papers	K1 - K9
17:15 - 18:30	Practical Performance Assessment 4 practical tasks in the Trading Simulator and TradingView (no coding): hypothesis testing, technical indicators and backtesting, ML model and A/B testing, and the multi-agent desk with forward test, diagnosis and report - covering A1 to A10.	All LOs	Individual assessment; Assessment papers	A1 - A10

Total instructional time, Day 2: 8 hours (480 minutes) = 6 hours training + 2 hours assessment, excluding the tea breaks and the 1-hour lunch break.

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7. Assessment Summary

Participants must be assessed as Competent (C) in BOTH assessment components to be awarded competency for this module. Each component is marked out of 100 against the marking points in the Assessment Rubric; Competent requires at least 80 marks in each component, at least 60% of the marks for every knowledge and ability statement, and every critical marking point awarded.

Method	Duration	Format	TSC Ref	Coverage
Written Assessment (Short Answer Questions)	45 minutes	9 open-ended short answer questions	K1 - K9	All 9 knowledge statements
Practical Performance	75 minutes	4 practical tasks (the Trading Simulator and TradingView, as in the labs - no coding; the Trading Simulator's paper broker or an Alpaca paper account) on a supplied dataset	A1 - A10	All 10 ability statements

Assessment format: Open book. Candidates may refer to their Learner Guide, the lab manuals and their own class notes. Refer to the Assessment Plan for the full assessment specification, marking guides and records.

8. Training Resources

Trainer Resources

- Trainer slide deck (PowerPoint), 228 slides covering all four topics and twenty labs.
- Lesson Plan (this document) and Facilitator Guide.
- Assessment Plan, including question papers, marking guides, rubric and Assessor Key.
- Lab folder (labs) with the twenty lab manuals; the Trading Simulator installed on the trainer laptop; a trainer Alpaca paper account for live demonstrations.
- Laptop with the Trading Simulator, a modern web browser, projector and whiteboard.

Learner Resources

- Learner Guide containing all topic content and step-by-step instructions for every lab.
- One laptop per participant with a modern web browser and internet access; a free Alpaca paper-trading account and a free TradingView account (created in Lab 1 and Lab 2).
- The lab folder with the twenty lab manuals (LabNN_Instructions.pdf).
- The course Trading Simulator (https://github.com/alfredang/algorithmic-trader-bot): a local web app (uv or Docker Compose) with a Course Labs page that walks through all twenty labs - no coding is needed.

9. Trainer Notes

- Paper trading only. Never demonstrate with a live-money account, and remind participants that nothing in the course is investment advice.

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- If the venue network blocks the Alpaca API, or a participant cannot open an account, every lab runs offline on the bundled data and the built-in paper-broker simulator. Show both paths early (Lab 1) so nobody is blocked.
- This is a low-code course: run every lab in the Trading Simulator (Course Labs page, quick and visual). No participant needs to write code. Participants who want full customisation in Python can be pointed to the separate Full Code edition. Have the simulator installed and seeded on the trainer machine before Day 1 so you can project each lab's page.
- Webhook alerts on TradingView need a paid plan. In Lab 17 demonstrate with the trainer's account if available, and let participants use the simulator's 'Run self-test (7 simulated alerts)', which sends the identical JSON a TradingView alert sends.
- Lab 18 contains deliberately seeded faults. Do not reveal them; let participants find the look-ahead first, because it produces no error message - only a result that is too good.
- Keep the finance context foremost. When a participant reports a great backtest, ask the three questions: is it net of costs, is it out of sample, and what would the investment committee need to see before risking capital?
- In Topic 1, link every milestone of the AI and algorithmic-trading histories to a control used later in the course (circuit breakers to the kill switch, Knight Capital to deployment testing, MiFID II to algorithm records).

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